

USRP B210 SDR Link — System Reference

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USRP B210 SDR Link — System Reference

End-to-end digital communication system for two USRP B210 radios (UHD), written in C++. This document lists **every feature, algorithm (with the math and the common alternatives), and command-line option** exposed by `sdr_system`.

- Binary: `phy/build/sdr_system`
- Help: `./sdr_system --help`
- Two operating styles: **one-way** (`--role tx / --role rx`, repeated sends, auto-terminating RX) and **ARQ** (`--role source_arq / --role sink_arq`, stop-and-wait with ACK).

1. Signal-chain overview

TX: message → split into chunks → CRC-16 → [FEC encode] → bits→symbols (modulator)
 ├─ single-carrier: RRC pulse-shape → preamble prepend → USRP TX
 └─ OFDM: QAM→IFFT+CP frame [SC-sync | chan-est | data] → USRP TX

RX: USRP RX → energy detector (burst gate) → AGC → frame/symbol sync
 ├─ single-carrier: matched filter → timing (Gardner) → [equalizer] → demod
 └─ OFDM: Schmidl-Cox sync + CFO → FFT → 1-tap eq → pilot CPE → demod
 → [FEC decode] → CRC-16 check → (ARQ: ACK if OK) → reassemble message

Default physical parameters: symbol rate 0.8 MHz, U/D = 2/1 → sample rate 1.6 MHz (integer **2 samples/symbol** on the wire), RRC roll-off 0.25, 151 taps.

2. Modulation

`--scheme <NAME>` selects the constellation. Supported (bits/symbol in parentheses):

Family	Schemes	bits/sym
PSK	BPSK (1), QPSK (2), 8-PSK (3)	1-3
QAM	16-QAM (4), 32-QAM (5), 64-QAM (6), 128-QAM (7), 256-QAM (8)	4-8
Differential	DBPSK (1), DQPSK (2), 8-DPSK (3), PI4-QPSK (2)	1-3
APSK	16APSK (4), 32APSK (5)	4-5

(16QAM and 16-QAM spellings both accepted.)

Math. A modulator maps $k = \log_2 M$ bits to one of M complex points $s = I + jQ$:

- **M-PSK**: points on the unit circle, $s = e^{j2\pi m/M}$, $m = 0, \dots, M - 1$. All symbols equal-energy; demod by nearest angle. Robust but spectrally inefficient at high order (points crowd on the circle).
- **M-QAM**: square/cross grid of amplitudes, $s = (2i - \sqrt{M} + 1) + j(2q - \sqrt{M} + 1)$. Best power efficiency for a given rate on an AWGN channel, but needs accurate amplitude -> sensitive to gain/EVM/clipping (this is why 16-QAM failed OTA at $\sim 20\%$ EVM while QPSK survived $\sim 37\%$).
- **APSK**: points on a few concentric rings. Lower PAPR than QAM, favored on non-linear (satellite) amplifiers — a middle ground between PSK and QAM.
- **Differential (D*)**: information is in the **phase change** between consecutive symbols, $s_n = s_{n-1} e^{j\Delta\varphi}$. No absolute phase reference needed -> tolerates carrier phase offset / no PLL, at a ~ 3 dB SNR penalty. PI4-QPSK rotates the constellation by $\pi/4$ each symbol to avoid zero-crossings (lower envelope variation).

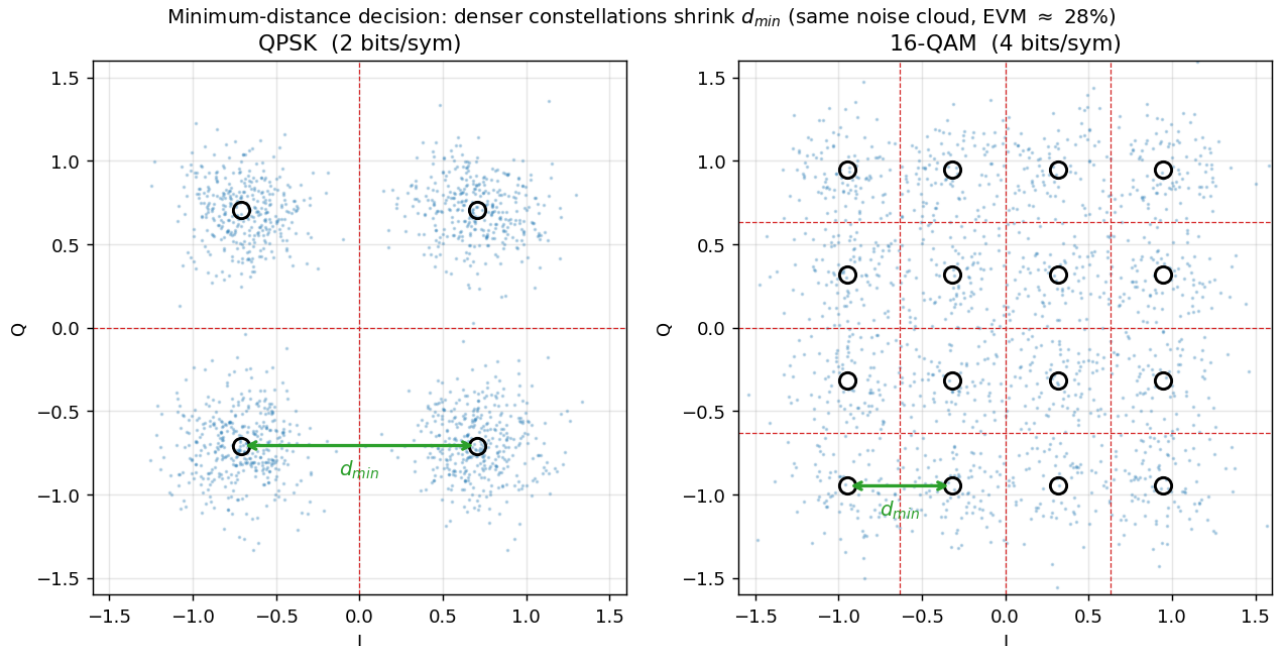
Detection & why constellation order costs SNR. The demodulator makes a **minimum-distance decision** — pick the constellation point nearest the received symbol y (optimal for equiprobable symbols in AWGN):

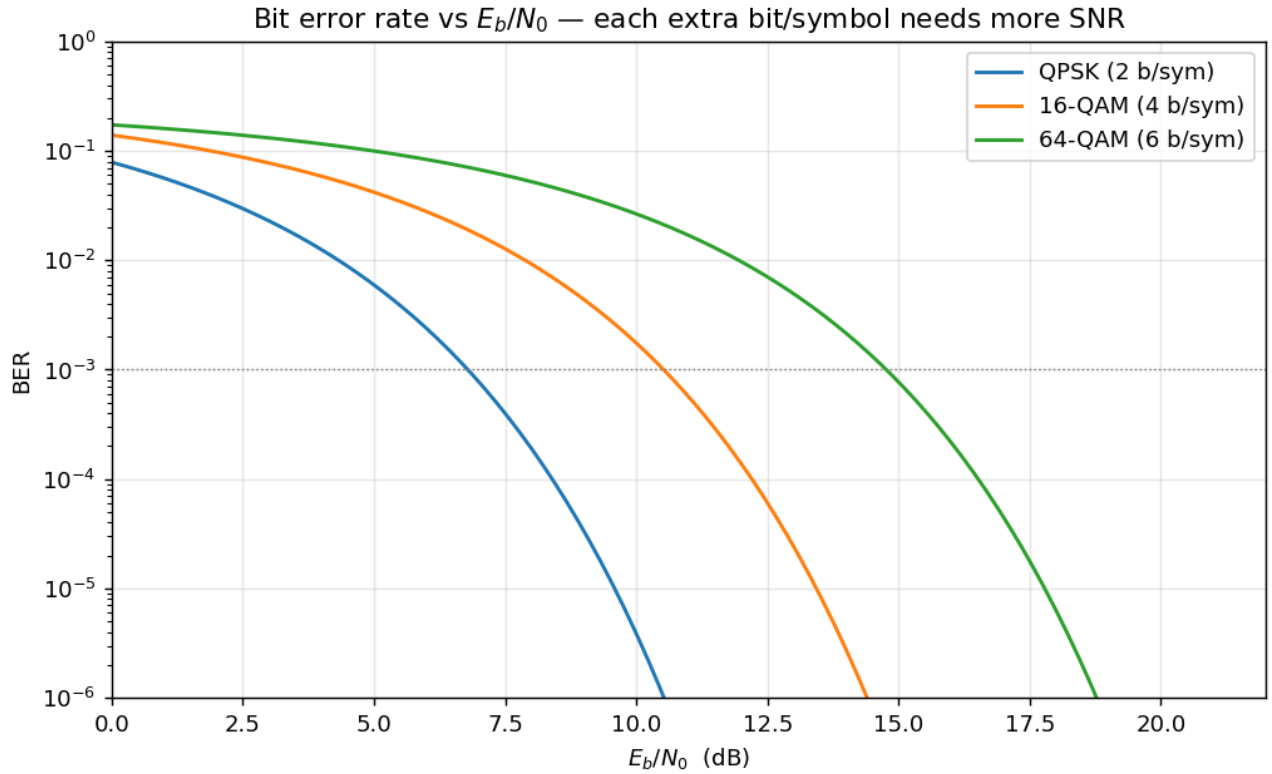
$$\hat{s} = \arg \min_{c \in \mathcal{C}} |y - c|$$

An error occurs when noise/distortion pushes y past the halfway line to a neighbour, i.e. beyond half the **minimum distance** $d_{\min}/2$. For a fixed average power E_s , packing more points shrinks $d_{\min}$: QPSK has $d_{\min} = \sqrt{2E_s}$ with the whole plane split into 4 quadrants, while 16-QAM squeezes 16 points into the same power so $d_{\min}$ is $\sim 3\times$ smaller and its decision cells $\sim 3\times$ tighter. The **error vector magnitude**

$$\text{EVM} = \frac{\text{rms}(y - \hat{s})}{\text{rms}(c)}$$

measures how far symbols land from ideal; reliable decoding needs roughly $\text{EVM} \lesssim d_{\min}/(2 \text{ rms})$, which is why QPSK survived $\sim 37\%$ EVM over the air but 16-QAM (needing $\lesssim 12\%$) did not. **Gray coding** the bit->point map (adjacent points differ by one bit) makes each symbol error cost only ~ 1 bit error.





Bits $\leftrightarrow$ symbols detail. Gray-like index mapping; `bits_to_index` zero-fills a partial final group so schemes whose bits/symbol don't divide the frame (32-QAM, 128-QAM) still align — a bug fixed early in this project.

Common alternatives not implemented: GMSK/CPM (constant-envelope, used in GSM/Bluetooth), non-square 8-QAM, probabilistic constellation shaping.

3. Waveform: single-carrier vs OFDM

`--waveform sc` (default) or `--waveform ofdm`.

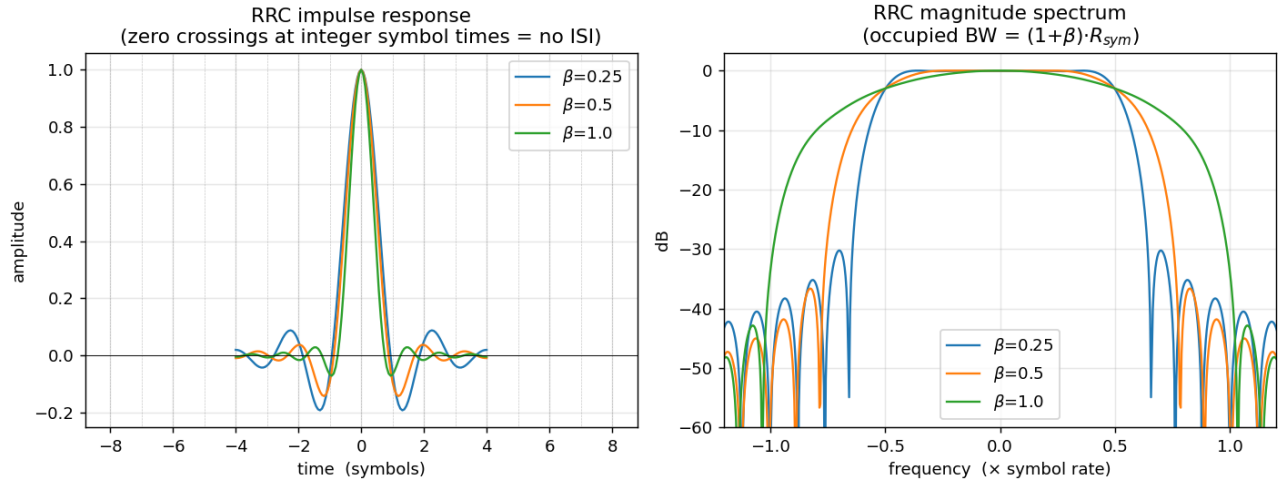
3.1 Single-carrier + RRC pulse shaping

Symbols are upsampled U/D and convolved with a **root-raised-cosine** filter (`--filter_type rrc`, `--roll_off 0.25`, `--num_taps 151`).

Math. A raised-cosine (RC) spectrum satisfies the **Nyquist ISI criterion** — its impulse response is zero at every other symbol instant, $g(mT) = \delta[m]$ — so neighbouring symbols don't interfere at the sampling times. Splitting it as **RRC at TX and RRC at RX** makes the cascade $G_{RRC}(f) G_{RRC}(f) = G_{RC}(f)$ a full RC (matched-filter optimal in AWGN) and maximizes SNR. The roll-off $\beta \in [0, 1]$ trades occupied bandwidth against sidelobe decay:

$$B = (1 + \beta) \frac{R_{sym}}{2} \quad (\text{one-sided baseband; occupied width} = (1 + \beta) R_{sym}).$$

Alternatives: `rc` (raised cosine, all shaping at TX), `lp` (plain low-pass); Gaussian shaping (GMSK) is the common constant-envelope alternative.



3.2 OFDM

Frame layout per burst: [Schmidl-Cox sync symbol | channel-estimation symbol | data symbols...], each symbol = N-point IFFT + cyclic prefix.

- `--ofdm-fft 64` — number of subcarriers N (FFT size).
- `--ofdm-cp 16` — cyclic-prefix length (must exceed the channel's delay spread).
- `--ofdm-tx-peak 0.5` — scales the frame so its peak $\approx$ this value (OFDM has high PAPR; keep the DAC out of clipping).

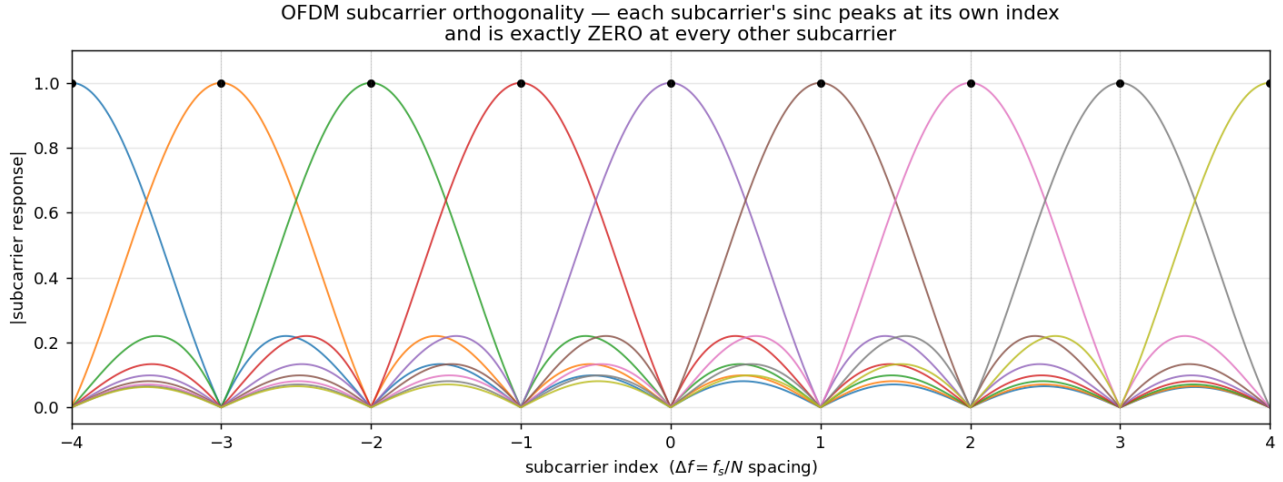
Math. OFDM splits the band into N narrow orthogonal subcarriers via the inverse DFT:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}.$$

A cyclic prefix (copy of the tail prepended) turns the channel's *linear* convolution into *circular* convolution, so after the FFT each subcarrier sees a single complex gain $H[k]$:

$$Y[k] = H[k] X[k] + W[k].$$

Equalization is then **one complex divide per subcarrier**, $\hat{X}[k] = Y[k]/H[k]$ — no FIR equalizer needed even under heavy multipath, provided the delay spread < CP. This is why OFDM is preferred for dense QAM on frequency-selective channels. Cost: high peak-to-average power ratio (PAPR) and sensitivity to carrier frequency offset (loss of subcarrier orthogonality $\rightarrow$ inter-carrier interference).



Channel estimation & pilots. One known **channel-estimation symbol** (known value on every active subcarrier) gives $H[k] = Y_{\text{chest}}[k]/\text{ref}[k]$. **Scattered pilots** (every 8th active subcarrier carries a known value) then track the residual **common phase error (CPE)** that grows symbol-by-symbol from residual CFO.

CPE estimation — the fix that made OFDM work OTA. Per data symbol, estimate the phase ϕ from the pilots and derotate:

$$\hat{\phi} = \arg\left(\sum_{k \in \text{pilots}} Y[k] H^*[k] \text{PILOT}^*\right) \quad (\text{maximum-ratio combining})$$

The earlier version used the *equalized* pilot $Y[k]/H[k]$, which **blows up on a deep-fade subcarrier** (tiny $|H[k]|$) — one garbage pilot then hijacked ϕ and smeared the whole symbol into a blob (EVM $\sim 67\%$, 0 CRC over the air; invisible in a mild simulation channel). Weighting by channel power $|H[k]|^2$ via $Y \cdot \text{conj}(H)$ suppresses faded pilots instead of amplifying them $\rightarrow$ four clean clusters (EVM $\sim 37\%$), message decodes. Disable with `env OFDM_NO_CPE=1` to compare.

Common alternatives: decision-feedback / per-subcarrier phase-locked tracking, MMSE channel estimation (vs the least-squares one used here), pilot-aided residual-CFO (SFO) slope correction, windowed-OFDM / filtered-OFDM / OTFS for spectral containment.

4. Energy detection (burst gating)

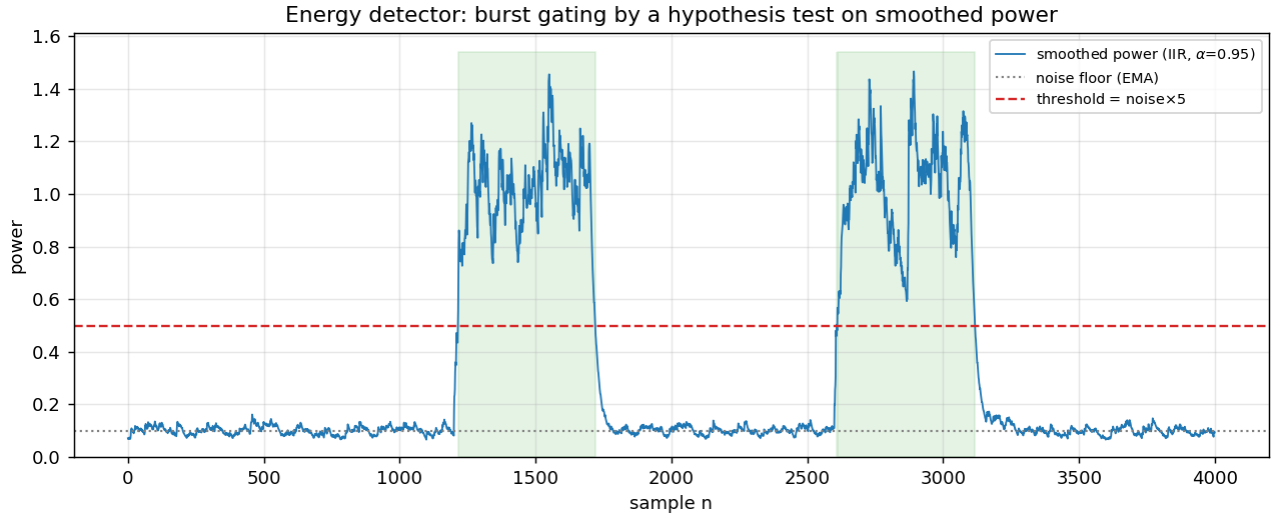
The RX runs continuously; the energy detector decides *when a burst is present* so downstream sync isn't run on noise.

Math. This is a **binary hypothesis test** per sample — H_0 : noise only, H_1 : signal present — on the received power. The instantaneous power $|r[n]|^2$ is smoothed by a one-pole IIR (exponential moving average) to reduce variance:

$$P[n] = (1 - \alpha) |r[n]|^2 + \alpha P[n - 1], \quad \text{declare burst if } P[n] > \gamma.$$

Larger α = more averaging (time constant $\approx 1/(1 - \alpha) \approx 20$ samples), which trades detection latency for a lower false-alarm rate — the smoothing collapses the variance of the noise-power estimate so a single threshold γ cleanly separates the two hypotheses. ($\alpha = 0.02$ barely smoothed $\rightarrow$ the detector fired on every noise spike and even chopped real bursts apart on the RRC envelope; $\alpha = 0.95$ gives one clean capture/burst.)

- **Adaptive threshold** (`--det-adaptive true`, default): $\gamma = \nu \cdot \hat{N}_0$ where $\hat{N}_0$ is the measured noise floor and $\nu = \text{--det-mult } 5$ (raise to 10–30 over the air so ambient RF doesn't trigger it; too high misses weak bursts).
- **Continuous noise tracking** (`--det-continuous true`, default): the noise floor itself is re-estimated by an EMA during idle periods, $\hat{N}_0 \leftarrow (1 - \beta)\hat{N}_0 + \beta |r[n]|^2$, so it follows drifting ambient noise. Alternative: one-shot startup calibration (`--det-continuous false`) — simpler but brittle if the environment changes.
- **Fixed threshold** (`--det-adaptive false + --det-threshold 1e-7`): absolute cutoff, only for a known clean cabled link.
- `--energy_packet_size` — how many samples to grab once a burst is detected (auto-sized per modulation).



Common alternatives: matched-filter / correlation detection (detect on the *known preamble* rather than energy — more sensitive but needs the template), constant-false-alarm-rate (CFAR) detectors, cyclostationary feature detection (works below the noise floor).

4.1 Channel sensing (occupancy / listen-before-talk)

The same energy measurement, used for a different decision: not “*is a burst starting now*” but “*is the channel occupied over this window*” — the basis for carrier-sense multiple access (CSMA). `--role sense` streams samples (no decode pipeline), integrates the average power over a window of `--sense-window ms`, and declares the band **busy** when it exceeds a threshold:

$$\bar{P} = \frac{1}{N} \sum_{n=0}^{N-1} |r[n]|^2, \quad N = f_s \cdot T_{\text{window}}, \quad \text{busy if } 10 \log_{10} \bar{P} > \gamma_{\text{dB}}.$$

It reports one machine-parseable line per window — `[SENSE] busy=.. power_db=.. peak_db=.. samples=..` — for `--sense-count` windows (**0 = stream forever**, a persistent feed). It is the same H_0/H_1 power test as §4, but integrated over a *fixed window* the caller chooses rather than gated to a burst envelope.

Calibration matters. The absolute power depends on `--rx-gain` and the ambient/DC-leakage floor, so γ_{dB} must be set *relative to the measured idle floor*, not hard-coded. The Python layer (`channel_sense.py`) auto-calibrates: it measures the idle floor over N windows and sets $\gamma_{\text{dB}} = \text{median}(P_{\text{idle, dB}}) + \text{margin}$ (default +6 dB). Validated on this rig: idle floor ≈ -12 dB, an active carrier ≈ -3 dB (a clean +9 dB separation), threshold auto-placed between them.

Reusable API (`channel_sense.py`, importable from any script):

- `sense_channel(...)` — one measurement -> {busy, power_db, ...} (re-inits the radio).
- `calibrate_floor(...)` — idle floor -> busy threshold.

- `SenseStream` — a **persistent** feed (`--sense-count 0`) with one radio init and a background reader that keeps the latest window fresh; `.should_transmit(p, thr)` is an instant decision with no per-call re-init (Python owns the threshold; the feed streams raw power_db).
- `should_transmit(p, ...)` — **p-persistent access**: if the channel is idle, transmit with probability p ; if busy, defer. The scaffold for CSMA/collision-avoidance on top of the PHY.

Frequency-range scan (`freq_scan.py`) sweeps a band to find a *quiet carrier* before a run: it retunes the RX across `--start/--stop` in `--step` MHz, senses the received power at each step (via `--role sense`), and prints a per-frequency table + an ASCII spectrum, ranking the quietest candidates (with an optional power-vs-frequency plot). E.g. `python3 freq_scan.py --start 902 --stop 928 --step 1 --rx-args addr=192.168.20.2`.

Common alternatives: full CSMA/CA with randomized exponential backoff, RTS/CTS handshaking, energy detection with hysteresis (separate busy/idle thresholds to avoid chattering), and — for sensing *below* the noise floor — matched-filter or cyclostationary detection as in §4.

5. Synchronization, frequency & phase offset

5.0 The received-signal model (what every block below is undoing)

After the RX front-end down-converts to baseband, the transmitted symbol stream $s[m]$ (pulse-shaped, sampled at f_s) arrives distorted by four separable impairments:

$$r[n] = \underbrace{A e^{j(2\pi \frac{\Delta f}{f_s} n + \varphi_0)}}_{\text{carrier}} \underbrace{\sum_m s[m] g(nT_s - mT - \tau)}_{\text{timing (delay } \tau)} + \underbrace{w[n]}_{\text{noise}}$$

- A — unknown gain -> removed by the **AGC** (§9).
- Δf — **carrier frequency offset (CFO)**: the two radios' oscillators differ (each B210 TCXO is ± 2 ppm, so up to ~ 4 ppm ≈ 3.6 kHz at 915 MHz). A constant Δf produces a phase that **grows linearly with time**, $\theta[n] = 2\pi(\Delta f/f_s)n$ — it *spins* the constellation. Left uncorrected it turns the constellation into a ring.
- φ_0 — **static carrier phase offset**: the oscillators' phase difference at acquisition plus the channel's phase. A constant *rotation* of the whole constellation.
- τ — **timing offset**: the ADC sampling grid isn't aligned to the symbol centers, so the matched filter is sampled off its peak -> inter-symbol interference (ISI).
- $w[n]$ — AWGN.

Note the coupling: **residual CFO becomes a phase ramp** $\varphi[n] = \varphi_0 + 2\pi(\Delta f/f_s)n$, which is why §5.5 uses a *second-order* loop (tracks a constant phase **and** its constant rate) and why OFDM needs per-symbol CPE tracking (§3.2). The recovery order is: **frame sync** -> **timing** -> **CFO** -> **phase**.

5.1 Preamble (the known reference all estimators key off)

`--preamble m-sequence` (default) or `--preamble zadoff`; `--m 5` sets the m-sequence order (length $L = 2^m - 1 = 31$).

A good preamble p has a **thumbtack autocorrelation** $\sum_n p[n] p^*[n+k] \approx E \delta[k]$ (large at zero lag, near-zero elsewhere) so the correlator (§5.2) gives one sharp, unambiguous peak.

- **m-sequence** — maximal-length binary (BPSK ± 1) shift-register sequence. Periodic autocorrelation is two-valued: L at zero lag, -1 otherwise. Real-valued.
- **Zadoff-Chu** — complex **CAZAC** sequence $p[n] = e^{-j\pi u n(n+1)/L}$: *constant amplitude* and *zero cyclic autocorrelation*. The flat spectrum makes it ideal for training a complex channel estimate / equalizer, and its constant envelope correlates cleanly even under CFO. Preferred for equalizer training and dense QAM.

For **differential** schemes the **last preamble symbol doubles as the differential reference**: the TX encodes the first data symbol relative to it, and the RX keeps that one symbol in front of the data so the differential decoder returns exactly N symbols (not $N - 1$). This keeps the decoded bit count aligned with the fixed FEC/CRC/ARQ

framing, and these schemes bypass the coherent phase PLL (§5.5) — a decision-directed loop would pin the absolute phase and destroy the transitions they carry. (The equalizer path prepends the *ideal* `preamble.back()` instead, since equalization has already normalized the data to the ideal frame.) Hardware-verified end-to-end for DBPSK / DQPSK / 8-DPSK; without this the reference off-by-one shifted the framing and every packet failed even though sync was perfect.

5.2 Frame synchronization — ACQ correlation

The receiver must find where the packet starts. `ACQSynchronizer::SamplesACQPerformance` slides the known preamble over the burst and computes, at each candidate offset τ (the `ComputeCorrelation` routine), the **matched-filter** / **cross-correlation** magnitude:

$$R(\tau) = \left| \sum_{n=0}^{L-1} p^*[n] r[\tau + n o_s] \right| \quad (o_s = \text{samples/symbol})$$

By the matched-filter theorem this is the optimal detector for a known sequence in AWGN. At the true start τ^* , all L terms add coherently $\rightarrow R(\tau^*) \approx \sum |p[n]|^2$; at any wrong offset the terms add with random phases $\rightarrow R \sim \sqrt{L}$ (much smaller). Detection:

- **Coarse pass:** find τ where $R(\tau)$ first crosses `--sync_threshold` (default 15), then
- **Fine pass:** search $\pm a$ few samples around it for the true maximum.
- **Noise floor / CFAR:** `EstimateNoiseFloor` correlates *away* from the peak (excluding a guard zone) to measure the off-peak floor, so the threshold can be set relative to it. Set `--sync_threshold` below the real peak ($\approx$ preamble length ~ 31 after AGC) but above that floor; watch the [ACQ] Peak correlation log lines.

Because the search tests **every sample** (not every o_s -th), the peak location simultaneously gives **coarse symbol timing** — the correct sub-symbol sampling phase. (Striding by o_s would test only one of the o_s phases and could lock a half-symbol off $\rightarrow$ catastrophic ISI, BER ≈ 0.5 . This was an actual bug; the brute-force search fixed it.)

5.3 Symbol-timing recovery — ACQ (active) vs Gardner TED (available)

What actually runs: ACQ, not Gardner. In packet mode the symbol timing is recovered *entirely* by the ACQ preamble correlation of §5.2 — because the correlator tests **every** sample offset, its peak already lands on the optimal sub-symbol sampling instant, and the aligned burst is extracted at one sample/symbol. No separate timing loop runs. This is a **feedforward, one-shot per-burst** estimate: fast, robust, and sufficient here because the sample-clock (SFO) drift across a single ~ 1000 -symbol burst is negligible (≈ 0.004 symbols at 4 ppm), so one timing phase holds for the whole burst.

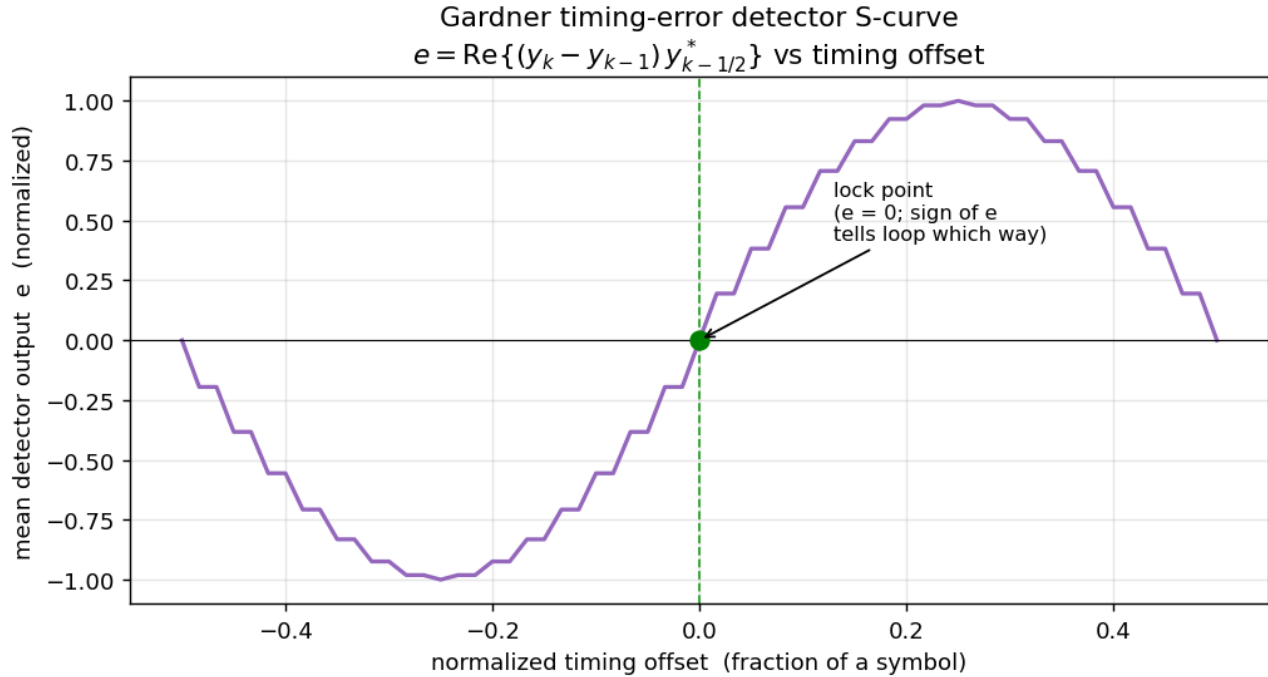
Available but dormant: the Gardner timing-error detector. `GardnerTED` and a ready `timing_recovery_thread` live in `timing_recovery.hpp`, and its knobs are exposed (`--timing_loop_bw 0.015`, `--timing_damping 0.707`), but it is **not wired into the current RX pipeline** — ACQ replaced it, and those two parameters are currently unconsumed (Gardner is exercised only by tests/`frontend_repro.cpp`). It is a *feedback* loop that tracks a continuously **drifting** fractional delay, which is needed only for a **true continuous symbol stream** (no per-burst preamble to re-sync on), where SFO accumulates over millions of symbols into whole-symbol drift. Drop it in between the matched filter and the CFO stage if/when such a waveform is added. The rest of this section describes that (available) detector.

Error detector. With two samples per symbol — one at the symbol center y_k and one at the half-symbol midpoint $y_{k-1/2}$ — the Gardner error is

$$e_k = \text{Re} \left\{ (y_k - y_{k-1/2}) y_{k-1/2}^* \right\}.$$

Intuition: if sampling is early or late, the midpoint sample $y_{k-1/2}$ (sitting on the transition between symbols) correlates with the slope $y_k - y_{k-1}$; at perfect timing the midpoint is a true zero-crossing and $e_k = 0$. It is **decision-independent and carrier-phase-independent** (the $\text{Re}\{\cdot\}$ with the difference term cancels a constant

rotation), so it works *before* CFO/phase are resolved — the classic reason Gardner is favoured for QAM *tracking* loops.



Loop. The NCO advances $\omega = 2/\text{sps}$ per input sample (two strobes/symbol — using $1/\text{sps}$ gives only one strobe and the "midpoint" lands a full symbol away -> the loop tracks garbage; this was a fixed bug). A **Farrow interpolator** produces the fractional-delay sample at offset μ , and a **proportional-integral (PI) loop filter** drives μ :

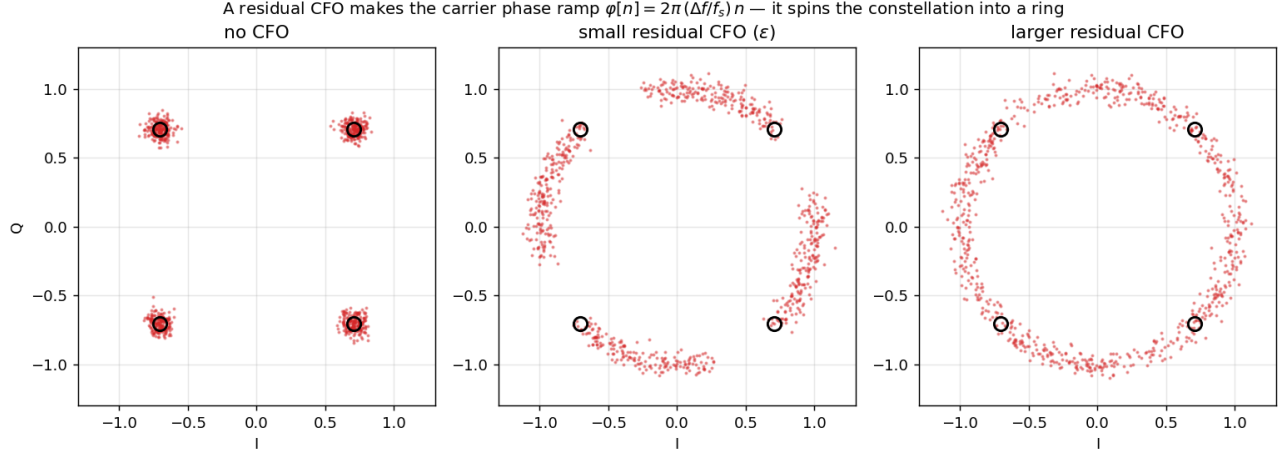
$$\nu \ +=\ K_2\ e_k \quad (\text{integrator — tracks a constant timing rate / clock offset}), \quad \mu \ -=\ K_1\ e_k \quad (\text{proportional}).$$

with ν clamped to $\pm 10\%$ of ω . The PI coefficients come from the standard second-order design (same θ_n/ζ formulas as §5.5). Only the symbol-center strobes are emitted -> exactly one sample per symbol out.

Alternatives: Mueller & Müller (decision-directed, needs only 1 sample/sym but needs carrier first), early-late gate, zero-crossing TED.

5.4 Carrier frequency offset (CFO) estimation & correction

A constant CFO spins the constellation at $2\pi \Delta f/f_s$ rad/sample. Three estimators are provided (CF0Estimator), all **data-aided** off the preamble, then FrequencyShifter removes it by multiplying sample n by $e^{-j2\pi(\Delta f/f_s)n}$ (a running phase accumulator, so blocks join seamlessly). The pipeline default is (C) **least-squares phase-slope**.



(A) Pilot-aided (phase-slope across preamble symbols). For preamble symbols $p[k]$ received at $r[k \cdot \text{sps}]$, the phase advance per symbol is measured with the known data stripped off:

$$\varphi_k = \arg(r[k \text{ sps}] r^*[(k-1)\text{sps}] p^*[k] p[k-1]), \quad \widehat{\Delta f} = \frac{f_s}{2\pi \text{ sps}} \overline{\varphi_k}.$$

Averaging over the preamble reduces noise. Unambiguous range $|\Delta f| < f_s/(2 \text{ sps})$ (one symbol of phase must not wrap).

(B) Auto-correlation (Moose / Schmidl-Cox). With a preamble made of two identical halves of length L , the second half equals the first times the CFO phase accrued over L samples:

$$P = \sum_{n=0}^{L-1} r[n] r^*[n+L] \approx |A|^2 L e^{j2\pi(\Delta f/f_s)L}, \quad \widehat{\Delta f} = \frac{f_s}{2\pi L} \arg(P).$$

Unambiguous range $|\Delta f| < f_s/(2L)$ — smaller $L \rightarrow$ wider capture range but noisier. This is the same estimator OFDM uses (§5.6), where $L = N/2$. Note (B) needs a **repeated** preamble; the single-carrier default preamble is a single m-sequence, so (B) is not used there.

(C) Least-squares phase-slope (default). Strip the modulation from every preamble symbol, $\theta_k = \arg(r[k \text{ sps}] p^*[k]) = \varphi_0 + 2\pi(\Delta f/f_s) \text{ sps } k + \text{noise}$, unwrap, and fit a magnitude-weighted straight line; the slope b gives $\widehat{\Delta f} = b f_s/(2\pi \text{ sps})$. Because it uses **all** preamble symbols jointly (vs (A)'s lag-1 differencing) this is the data-aided ML / CRLB estimate — same $\pm f_s/(2 \text{ sps})$ range as (A) but $\sim L^2$ **lower variance**. A hardware-free Monte-Carlo (length-31 m-seq) puts the CFO-estimate RMSE at ~ 450 Hz vs ~ 1050 Hz for (A) at 12 dB symbol SNR — i.e. it roughly **halves** the ± 1200 Hz jitter of §13 — and it wins at every SNR the link actually decodes at (both fail near 0 dB, where phase unwrap breaks down). A **cross-burst EMA prior** (`--cfo_prior_alpha`, default 1.0 = per-burst) can further smooth the estimate for a *warm resident LO*; keep it at 1.0 for a cold per-fire LO (two-host DQPSK), whose CFO changes every burst.

Why residual CFO still matters. $\arg(\cdot)$ limits the estimate; whatever is left is a slow phase ramp the **phase tracker** (§5.5) or **OFDM pilot CPE** (§3.2) cleans up. Getting this balance wrong is exactly what smeared 16-QAM/OFDM over the air.

5.5 Carrier phase offset — estimation & PLL tracking

After timing + CFO the symbols still carry a residual phase $\varphi[n] = \varphi_0 + (\text{residual ramp})$. For **absolute** constellations (BPSK/QPSK/QAM) all of it must be removed before hard decisions; for **differential** schemes the static φ_0 cancels automatically (demod uses phase *differences*).

One-shot estimators (PhaseOffsetEstimator):

- **ML / preamble correlation** — optimal given the known preamble: $\hat{\varphi} = \arg(\sum_n r[n] p^*[n])$.
- **M-th power (blind, for M-PSK)** — raising to the M -th power collapses all M data phases onto one, removing the modulation: $\hat{\varphi} = \frac{1}{M} \arg(\sum_n r[n]^M)$.
- **Decision-directed** — $\hat{\varphi} = \arg(\sum_n r[n] \hat{s}^*[n])$ against the nearest constellation points $\hat{s}$ (used for QAM, after a coarse lock).

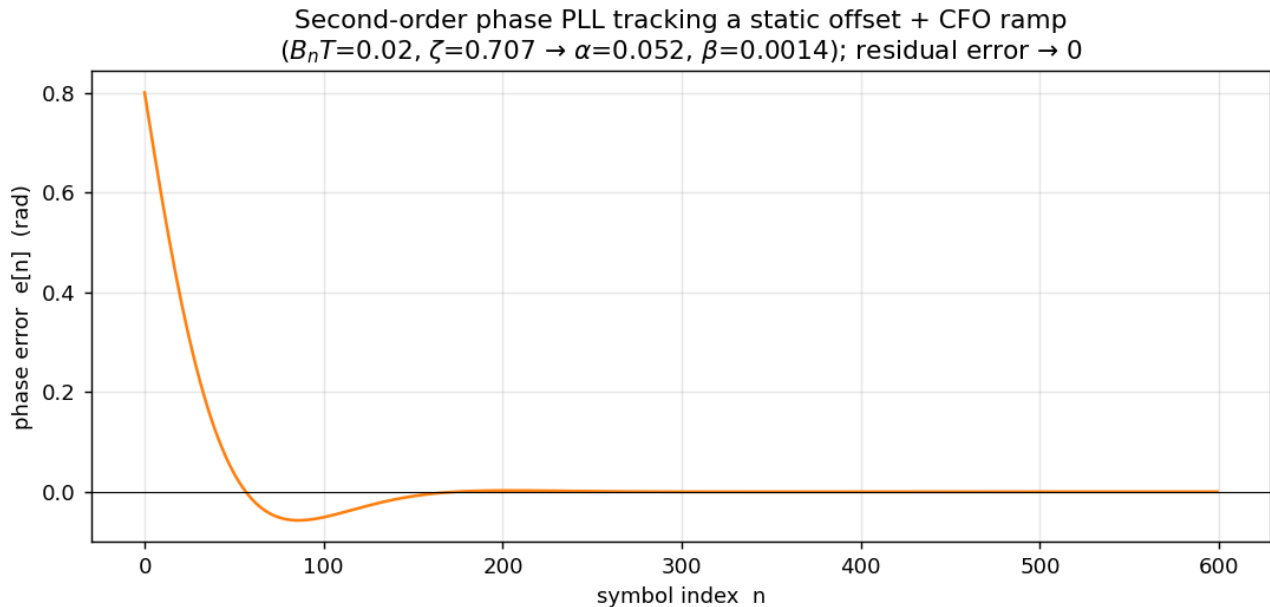
Tracking PLL (PhaseTracker) — a second-order digital PLL that follows a static offset **and** a constant frequency ramp (residual CFO). Per symbol: form the phase error against the nearest decision, then update an integrator (frequency) and the phase:

$$\begin{aligned} e[n] &= \arg(\text{corrected}[n] \hat{s}^*[n]) && \text{(phase detector, angle in } [-\pi, \pi]) \\ \text{freq} &+= \beta e[n] && \text{(integrator — tracks the residual CFO ramp)} \\ \varphi &+= \alpha e[n] + \text{freq} && \text{(NCO phase)} \end{aligned}$$

The loop coefficients come from the standard proportional-integral design for loop bandwidth $B_n T$ (--phase_loop_bw 0.02) and damping ζ (--phase_damping 0.707):

$$\theta_n = \frac{B_n T}{\zeta + 1/(4\zeta)}, \quad \alpha = \frac{4\zeta \theta_n}{1 + 2\zeta \theta_n + \theta_n^2}, \quad \beta = \frac{4\theta_n^2}{1 + 2\zeta \theta_n + \theta_n^2}.$$

($\beta = 0$ gives a first-order loop.) Smaller $B_n T$ = less jitter but slower pull-in; $\zeta = 0.707$ is the critically-damped standard.



Alternatives: Costas loop (joint carrier recovery on the raw samples), block Viterbi-Viterbi phase estimation, pilot-symbol-aided interpolation.

5.6 OFDM synchronization — Schmidl & Cox

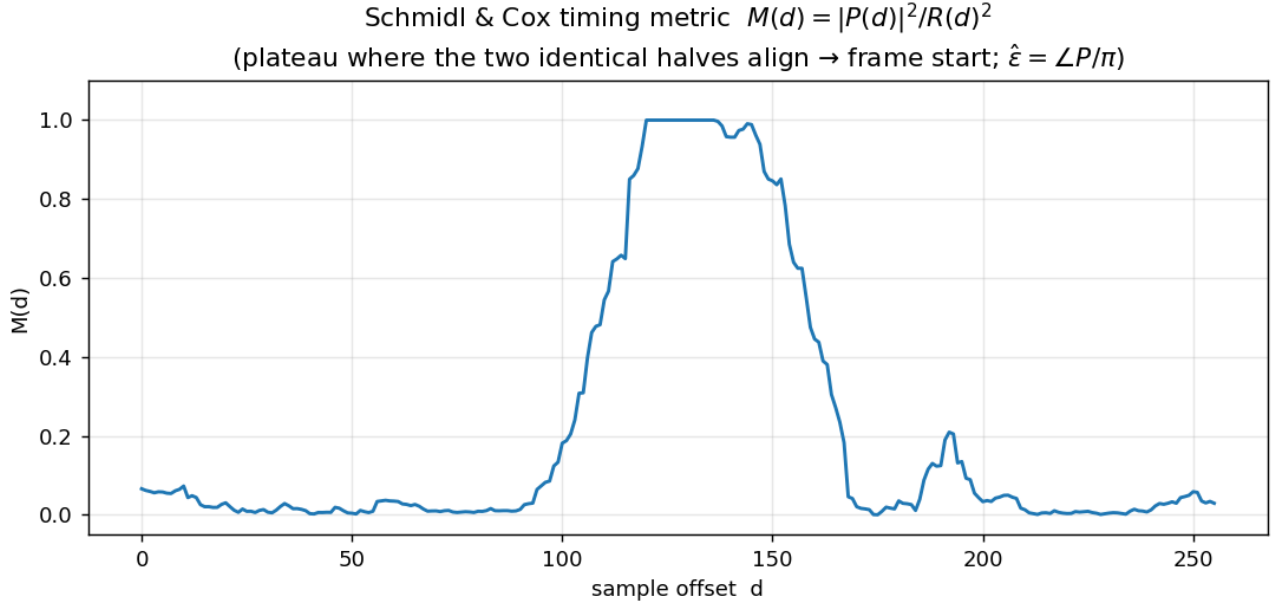
OFDM recovers frame timing and CFO jointly from a sync symbol built as two identical time halves [A A] (generated by loading only the **even** subcarriers). Sliding a window of length $N/2$:

$$\begin{aligned}
P(d) &= \sum_{n=0}^{N/2-1} r[d+n] r^*[d+n+N/2] \quad (\text{half-to-half correlation}) \\
R(d) &= \sum_{n=0}^{N/2-1} |r[d+n+N/2]|^2 \quad (\text{energy normaliser}) \\
M(d) &= |P(d)|^2 / R(d)^2 \quad (\text{timing metric } \in [0, 1])
\end{aligned}$$

$M(d)$ forms a **plateau** where the two halves align; the frame start is the plateau's left edge plus a small guard into the CP (any offset inside the CP is just a per-subcarrier phase the channel estimate absorbs). The **fractional CFO** falls out of the same correlation:

$$\hat{\epsilon} = \arg(P) / \pi \quad (\text{subcarrier-spacing units, range } \pm 1 \text{ subcarrier}).$$

and the burst is derotated by $e^{-j2\pi\epsilon n/N}$. Residual CFO / phase drift across the frame is then tracked per symbol by the scattered-pilot CPE of §3.2. An energy gate ($R \geq 0.30 \max R$) stops the metric from locking onto the low-power noise pad before the burst.



Alternatives: Minn and Park metrics (sharper, no plateau ambiguity), CP-based blind sync.

6. Channel equalization (single-carrier)

--eq_type None (default) | LMS | RLS | DFE. --eq_taps 11, --eq_mu 0.3 (NLMS step), --eq_dd false (decision-directed tracking after training).

Math. A length- T linear FIR equalizer w estimates each symbol from a window of received samples, $\hat{s}[k] = w^H r[k] = \sum_i w_i^* r[k-i]$, choosing w to invert the channel h (ideally $w * h \approx \delta$). Two ways to find w :

- **Least-squares / MMSE training** (used, exact for the complex Zadoff-Chu preamble). Stack the preamble windows into a matrix R and the known symbols into s ; minimizing $\|Rw - s\|^2$ gives the normal equations

$$w = (R^H R + \lambda I)^{-1} R^H s$$

solved here by Gaussian elimination. λI = Tikhonov / diagonal-loading regularization: it bounds noise enhancement at channel nulls (the MMSE, not zero-forcing, solution).

- **NLMS adaptive update** (real preamble / decision-directed tracking): step down the instantaneous-error gradient, normalized by input power for stability:

$$e[k] = s[k] - \hat{s}[k], \quad w \leftarrow w + \mu \frac{e^*[k] r[k]}{\|r[k]\|^2 + \varepsilon} \quad (\text{--eq_mu } 0.3, 0 < \mu < 2).$$

The LS solution places the main tap at the equalizer center, so a **center-tap group delay** of $(T - 1)/2$ samples must be compensated when aligning the output — mishandling this looked like "divergence" and was a fixed bug.

Status / why default None. On a clean cabled link no equalizer is needed. The LMS decision-directed loop currently **diverges on the real hardware signal** (DD error grows and destroys the symbols) — verified OTA where None decodes and LMS produces garbage. For frequency-selective channels, prefer **OFDM** (§3.2), which equalizes per-subcarrier without an adaptive FIR. Alternatives: DFE (feedback of past decisions, better on deep nulls), RLS (faster convergence than LMS, $O(n^2)$ cost), MLSE/Viterbi equalization (optimal, expensive).

7. Forward error correction (FEC)

`--fec true` (default off) — rate-1/2, constraint-length-7 **convolutional code** with **Viterbi** hard-decision decoding. Must match on both ends; halves the payload rate ($2\times$ symbols).

Math. Encoder: a 6-stage shift register holds the last $K - 1 = 6$ input bits (state $\sigma \in \{0, \dots, 63\}$); each input bit u emits two coded bits as mod-2 (XOR) taps of the register per the generator polynomials $G_1 = 171_8$, $G_2 = 133_8$ (the standard NASA / 802.11a code):

$$c_1 = G_1 \cdot [u, \sigma] \bmod 2, \quad c_2 = G_2 \cdot [u, \sigma] \bmod 2.$$

Each input bit thus influences $K = 7$ output-bit pairs (the constraint length / memory), which is what lets the decoder recover it even when some of those bits are flipped.

Viterbi decoding finds the maximum-likelihood transmitted sequence by the most-likely path through the 64-state trellis. For hard decisions the branch metric is the **Hamming distance** between the received pair and each branch's expected (c_1, c_2) ; the decoder keeps, per state, the survivor path with the smallest cumulative metric:

$$\text{PM}_t(\sigma') = \min_{\sigma \rightarrow \sigma'} \left[\text{PM}_{t-1}(\sigma) + d_H(r_t, c(\sigma \rightarrow \sigma')) \right]$$

then traces the survivors back to output the bits. It corrects any error pattern up to roughly $\lfloor (d_{\text{free}} - 1)/2 \rfloor$ — $d_{\text{free}} = 10$ for this code — per window; validated $\sim 100\%$ CRC-OK up to $\sim 2\%$ raw BER. (Two original bugs — encoder emitting from the *next* state, traceback storing the bit instead of the predecessor state — were fixed.)

Soft-decision Viterbi (Euclidean/LLR branch metric instead of Hamming) is now implemented via `--fec_soft` — ~ 2 dB over the hard-decision decoder (see §7.4).

The FEC is now **pluggable** — `--fec-type {conv, ldpc, turbo}` selects one of three rate-1/2 codes. The convolutional code above is the default; the sections below cover the two added near-Shannon codes and the shared selector/tuning and soft-decision layer.

7.1 Code-family selector and tuning

`--fec-type` (needs `--fec true`) picks the code; all three are rate 1/2, so the rest of the pipeline (symbol sizing, ARQ, CRC) is unchanged, and both ends must select the same code. The convenience layer in `fec.hpp` dispatches `fec_encoded_len` / `fec_encode_block` / `fec_decode_block` / `fec_soft_decode_block` to the chosen

codec; `fec_set_type()` builds it once at startup and `fec_set_tuning()` applies the knobs. The block codes (LDPC, turbo) segment the packet into k -bit blocks (`--ldpc-k`, default 256; last block zero-padded) and encode each to $2k$ coded bits.

Tuning knobs:

- `--fec-iters N` — max decoder iterations (LDPC belief-propagation / turbo BCJR; $\emptyset$ = default 50 / 6). Raise (e.g. turbo 8–12) if a marginal link won't converge. Decoder-only — need not match.
- `--fec-scale F` — normalized min-sum (LDPC) / extrinsic (turbo) scale, 0.7–0.9 typical ($\emptyset$ = default 0.75). Decoder-only.
- `--ldpc-k K` — block size (LDPC **and** turbo). Larger = stronger, more latency/padding. **Must match both ends.**
- `--ldpc-col-weight W` — LDPC variable-node degree (default 3); higher = denser code. **Changes the parity-check matrix — must match both ends** (turbo ignores it).

7.2 LDPC (`ldpc.hpp`)

Rate-1/2 systematic **IRA / staircase** LDPC. The parity-check matrix $H = [H_u \mid H_p]$ is built deterministically from a fixed seed (identical on both ends): the information part H_u places `col_weight` ones per info column, balanced across the $m = k$ check rows; the parity part H_p is lower-bidiagonal (an accumulator). That structure gives $O(n)$ **systematic encoding** — the parity is a prefix-XOR of the info syndrome, $p_r = \bigoplus_{i \leq r} s_i$ with $s = H_u u$ — and parity variable nodes of degree 2 (a genuinely good short code, unlike the degree-1 parity of an $H = [P \mid I]$ construction).

Decoding is flooding **normalized min-sum** belief propagation on the Tanner graph. Each iteration: every variable node sums its channel LLR and the incoming check messages; every check node returns $\text{sign} = \prod \text{sgn}$ and $\text{magnitude} = \alpha \cdot \min |L|$ over its *other* edges (attenuation $\alpha = \text{--fec-scale}$, default 0.75). It **early-terminates** the moment the hard-decision syndrome $H\hat{x} = 0$ clears, so decode cost falls as SNR rises. LLR convention matches `soft_demodulate_llr` (positive = bit 0).

7.3 Turbo (`turbo.hpp`)

Rate-1/2 **punctured parallel-concatenated convolutional code** (PCCC): two identical (7,5) recursive-systematic convolutional (RSC) encoders — one on the info bits, one on an interleaved copy (deterministic interleaver) — with the two parity streams punctured (alternate bits) to reach rate 1/2. The RSC has feedback $g_0 = 1 + D + D^2$ (7₈) and feedforward $g_1 = 1 + D^2$ (5₈), memory 2 -> a **4-state** trellis.

Decoding is the classic iterative turbo decoder: two soft-in/soft-out (SISO) **max-log-MAP (BCJR)** decoders exchange **extrinsic** LLRs through the interleaver for a few iterations, then take a hard decision on the combined LLR. Each SISO runs forward (α) and backward (β) recursions over the trellis with branch metric $\gamma \propto \frac{1}{2}(x_u(L_a + L_s) + x_p L_p)$, forms $L(u) = \max_{u=1}^* (\alpha + \gamma + \beta) - \max_{u=0}^* (\alpha + \gamma + \beta)$, and passes the scaled extrinsic $L_e = \text{scale} \cdot (L - L_a - L_s)$ to the other decoder (deinterleaved). Turbo has the **steepest waterfall** of the three codes (best coding gain per dB), and like LDPC it early-terminates as the link improves.

7.4 Soft-decision decoding

`--fec_soft` (RX-side) feeds per-bit **LLRs** from `soft_demodulate_llr()` (a max-log demapper on the equalized symbols; positive = bit 0) to the decoder instead of hard bits:

- **Convolutional:** soft Viterbi (Euclidean/LLR branch metric) — ~ 2 dB over hard.
- **LDPC / turbo:** soft is their *native* input; hard-decision (mapping bits to $\pm$ LLR) throws away most of their gain.

The single-carrier RX emits LLRs from the phase-corrected symbols. **OFDM** now does too: `ofdm_demodulation_thread` computes them from the per-subcarrier-equalized symbols (with a decision-directed noise-variance estimate) and feeds `rx_llr_fifo`. Previously the OFDM path produced only hard bits, so `--fec_soft` on OFDM silently fell back to hard decision — a large loss for the iterative codes.

Comparison (hardware-free, QPSK). At 2 dB E_b/N_0 , decode success is turbo-soft $\approx 90\%$ $>$ LDPC-soft $\approx 75\%$ $>$ conv-soft $\approx 52\%$. Decode cost: Viterbi is fixed (it walks the full 64-state trellis every packet), whereas LDPC (min-sum) and turbo (BCJR) early-terminate, so their cost drops as SNR rises. Ranking at short block sizes: turbo = best gain (moderate cost), LDPC = fastest with good gain, conv = weakest gain but predictable, bounded latency.

Common alternatives not implemented: Reed-Solomon (burst errors), Polar codes (5G control).

8. Error detection + ARQ

- **CRC-16-CCITT** appended to every chunk; the RX drops any chunk that fails the check (guarantees the delivered message is error-free).
- **Stop-and-wait ARQ** (`--role source_arq / sink_arq`): the source sends a chunk and waits for an ACK; on `--timeout 3000` ms it retransmits; it advances when ACKed and exits when all chunks are ACKed. The sink CRC-verifies each chunk, ACKs only verified ones, re-ACKs duplicates, and auto-stops once every chunk is in.
- **ACK transport** (`--ack-transport`):
 - `tcp` (default) — ACK over a socket; **no reverse RF path needed**. Source connects to `--ack-host 127.0.0.1 : --ack-port 5599` (the sink listens there). Ideal when both radios share one host.
 - `rf` — ACK sent back over the air on the second RF path (needs a clean reverse link).

Math/theory. Treat the message bits as coefficients of a polynomial $M(x)$ over GF(2); append 16 zeros ($M(x) \cdot x^{16}$) and divide by the CRC-16-CCITT generator $G(x) = x^{16} + x^{12} + x^5 + 1$ (mod-2 / XOR long division). The 16-bit remainder is the CRC; the RX recomputes it and flags any nonzero mismatch. Because an undetected error requires the error polynomial $E(x)$ to be an exact multiple of $G(x)$, this structure guarantees detection of: all single- and double-bit errors, any odd number of errors (G has the factor $x + 1$), and all burst errors of length ≤ 16 . Residual undetected probability $\approx 2^{-16}$ for random errors.

Stop-and-wait is the simplest ARQ (throughput limited by idling one round-trip per chunk). *Alternatives:* Go-Back-N / Selective-Repeat (pipelined, higher throughput), Hybrid-ARQ (FEC + ARQ combined — what you get here with `--fec on`: FEC corrects most errors, CRC+retransmit catches the rest).

One-way mode (`--role tx / rx`, no ACK): the TX cycles all chunks `--tx-reps` times; the RX reassembles from CRC-verified chunks and auto-stops `--rx-idle-timeout 8` s after the last burst.

8.1 Measuring the real BER (`--ber-expected`)

CRC is a **binary** verdict — a failed frame has ≥ 1 error, but CRC can't say whether that's one bit or hundreds. To quantify it, give the sink the **known transmitted payload** (`--ber-expected <file>`) and it compares the decoded bits to ground truth for **every** burst (CRC pass or fail), printing two rates:

- **pre-FEC BER** — raw channel bit errors (demodulated bits vs the transmitted *coded* bits). This is the true channel quality, before any correction.
- **post-FEC payload BER** — residual errors after Viterbi decoding (decoded payload vs the known payload). This is what the CRC actually sees; non-zero here $\Rightarrow$ CRC fails.

$$\text{BER} = \frac{\#\{\text{bits that flipped}\}}{\#\text{bits}}.$$

So a CRC-fail frame is **not necessarily garbage** — in a *marginal* window you see e.g. pre-FEC 1.2%, post-FEC 0.3% (nearly right, CRC catching 1–2 residual bits). The key diagnostic signature is **post-FEC $>$ pre-FEC**: once the raw BER exceeds the rate- $\frac{1}{2}$ K=7 code's decoding threshold (~ 10 – 11%), the Viterbi decoder chooses a wrong trellis path and **amplifies** errors instead of correcting them (*catastrophic FEC failure*) — measured on a broken link as pre-FEC 24%, post-FEC 42% ($\approx$ random $\Rightarrow$ genuine garbage). The processing gain of the preamble correlation means a burst can **detect strongly yet decode to garbage** (detection $\neq$ decodability).

Python: `marl_phy.ber_probe(n, scheme=...)` (or `python3 marl_phy.py ber`) runs a warm AP with the known payload, fires N copies, and reports min/median/max of both BERs — a quick link-quality meter at any distance.

9. Automatic gain control (AGC)

--AGC_type Feed (feed-forward, default) or Closed (closed-loop). Feed-forward measures the detected burst's RMS and scales it to unit power in one shot (no loop lag); closed-loop adapts a gain over time. The energy detector's captured burst feeds the AGC before sync.

10. Visualization / EVM

--viz true (default) captures one TX and one RX burst and **auto-saves a figure on exit** to `<viz-dir>/<scheme>/figure.png` (per-modulation folder, --viz-dir viz). The 2×3 figure shows TX/RX **time domain**, **spectrum** (FFT), and **constellation** with the ideal points overlaid and an **EVM %** readout. Rough EVM ceilings for reliable decode: QPSK tolerates $\sim 30\text{--}35\%$ (more with FEC), 16-QAM needs $< \sim 12\%$, 64-QAM $< \sim 6\%$. Render manually with `python3 phy/tools/plot_viz.py <dir> --fs 1.6e6 --save out.png`.

10.1 Why a messy constellation can still decode perfectly

A common and important surprise: the saved RX constellation can look like a smeared ring — the clusters visibly overlapping — yet the message decodes with zero errors. This is not a contradiction; it is the whole point of the error-control stack. Four facts reconcile it.

1. The plot is a single snapshot, not the data that decoded. viz.hpp writes each figure **once, from the first captured burst of the run** (written once / "only the FIRST time called"). The message, by contrast, is assembled from *many* bursts (§8). So the picture you are staring at is one — possibly poor — burst, while the message may have been built from later, cleaner bursts you never see plotted. The snapshot over-represents the worst moment.

2. "Messy" is not "random." A *truly uniform* circle — phase spread evenly over $[0, 2\pi)$ — genuinely cannot be decoded: for M -PSK it gives an $\approx (M - 1)/M$ symbol-error rate, far past any code's reach. The fact that decoding *succeeds* is itself proof the constellation is **not** uniform: the M clusters are spread and overlapping but still statistically centred on the right points, so **most** symbol decisions are still correct (often 80–90%). The eye exaggerates the mess — a scatter of thousands of alpha-blended points plus a few outliers reads as a ring even when the modes dominate.

3. Three layers turn "mostly right" into "exactly right." The decision slicer's raw output is only the first stage:

$$\text{symbol decisions (some wrong)} \rightarrow \underbrace{\text{Viterbi FEC}}_{\text{rate-1/2, } K=7} \rightarrow \underbrace{\text{CRC-16}}_{\text{detect residual}} \rightarrow \underbrace{\text{ARQ / --tx-reps}}_{\text{retransmit}}$$

FEC (§7) corrects a large fraction of the bit errors; CRC (§8) rejects any chunk that still contains one; ARQ / repetition then retransmits until a clean copy lands. The last layer is decisive: **you need only ONE CRC-clean copy of each chunk**. A link that fails 80% of bursts still delivers a perfect message — the passing 20% are kept, the rest dropped. The message is built from the *lucky* bursts, not the average one.

4. Two different quantities. The constellation/EVM measures the **raw, uncoded** channel quality — how hard the demod's job is. The decoded message reports what survives **after** the coding layers. They are *allowed* to disagree, and the gap between them is exactly the margin that FEC + CRC + retransmission buys:

Quantity	What it reflects
Constellation / EVM	raw link quality (pre-coding)

Quantity	What it reflects
Decoded message	result after FEC + CRC + ARQ

Note the 8-PSK mapping here is *natural binary*, **not** Gray-coded (`modulator.cpp` `create_8psk_constellation`), so none of this robustness comes from a benign bit-assignment — the coding layers do all of the work. Switching to Gray coding (adjacent symbols differing by one bit) would widen this margin further, a cheap future improvement.

This is also the precise mechanism behind the dense-QAM wall (§13): there, even a *single* clean copy never lands — the raw symbol errors exceed what FEC can carry and no burst passes CRC — so retransmission has nothing good to keep. 8-PSK sits right at the edge where the raw symbols are borderline but the stack still closes the gap; dense QAM/PSK is past it.

11. Complete command-line reference

Config file. Every option below can be set in a file instead of on the command line: `--config phy.cfg` reads `name = value` lines (# comments; long option name without `--`), and anything on the command line **overrides** the file — so you keep one edited config and tweak per-run. `phy.cfg` is a fully-defaulted template auto-generated on build (`phy/tools/gen_config_template.py`) from `sdr_system --help`, so it never drifts from the code.

Mode / roles

Option	Default	Controls
<code>--role</code>	<code>both</code>	<code>tx</code> , <code>rx</code> , <code>both</code> , <code>source_arq</code> , <code>sink_arq</code>
<code>--mode</code>	<code>source</code>	legacy source/sink selector
<code>--tx-reps</code>	<code>20</code>	one-way: times to cycle all chunks (no ACK)
<code>--rx-idle-timeout</code>	<code>8.0</code>	one-way RX: auto-stop after N s of no bursts (0 = until Ctrl-C)
<code>--num_bits</code>	<code>1000</code>	payload bits per packet
<code>--interval</code>	<code>3000</code>	TX gap between packets (ms)

ARQ / ACK

Option	Default	Controls
<code>--ack-transport</code>	<code>tcp</code>	ACK channel: <code>tcp</code> or <code>rf</code>
<code>--ack-host</code>	<code>127.0.0.1</code>	TCP: sink IP the source connects to
<code>--ack-port</code>	<code>5599</code>	TCP: ACK socket port
<code>--timeout</code>	<code>3000</code>	source ACK timeout (ms) before retransmit
<code>--timer_interval</code>	<code>1000</code>	sink ACK timer interval (ms)

Modulation / waveform / FEC

Option	Default	Controls
<code>--scheme</code>	<code>QPSK</code>	constellation (§2)
<code>--fec</code>	<code>false</code>	enable FEC (must match both ends)
<code>--fec-type</code>	<code>conv</code>	<code>conv</code> / <code>ldpc</code> / <code>turbo</code> (§7); must match both ends
<code>--fec_soft</code>	<code>off</code>	RX soft-decision (LLR) decoding (§7.4); coherent schemes
<code>--ldpc-k</code>	<code>256</code>	LDPC/turbo info-block size (bits); must match both ends
<code>--fec-iters</code>	<code>0</code>	max decoder iterations (0 = default 50/6); decoder-only

Option	Default	Controls
--fec-scale	0	min-sum/extrinsic scale (0 = 0.75); decoder-only
--ldpc-col-weight	3	LDPC variable-node degree; must match both ends
--waveform	sc	sc (single-carrier) or ofdm
--ofdm-fft	64	OFDM subcarrier count (FFT size)
--ofdm-cp	16	OFDM cyclic-prefix length
--ofdm-tx-peak	0.5	OFDM peak scaling (PAPR / clipping guard)
--sps	2	samples/symbol (informational)

Preamble / pulse shaping

Option	Default	Controls
--preamble	m-sequence	m-sequence or zadoff
--m	5	m-sequence order (length $2^m - 1$)
--add_preamble	true	prepend preamble to each packet
--filter_type	rrc	rrc / rc / lp
--roll_off	0.25	RRC/RC roll-off β
--num_taps	151	filter tap count
--U / --D	2 / 1	pulse-shaper up/down sampling (wire sps = U/D)
--symbol_rate	0.8e6	symbol rate (Hz)
--num_threads	1	FFT threads

Energy detection

Option	Default	Controls
--alpha	0.95	IIR power-smoothing (larger = smoother)
--det-adaptive (--IIR_threshold_adaptive)	true	use noise_floor $\times$ mult vs fixed threshold
--det-mult (--IIR_threshold_multiplier)	5.0	adaptive threshold multiplier (raise OTA to 10–30)
--det-threshold (--energy_threshold)	1e-7	fixed threshold (adaptive off only)
--det-continuous	true	re-track noise floor during idle vs one-shot calib
--energy_packet_size	3300	samples to collect after detection (auto-sized)
--IIR_window_size	20	IIR window size

Synchronization / timing / phase

Option	Default	Controls
--sync-threshold (--sync_threshold)	15.0	ACQ correlation gate (set below true peak, above noise)
--sps_sync	5	samples/symbol at matched-filter output
--recv_msg_len	auto	data symbols to extract per burst (auto from scheme)
--timing_loop_bw	0.015	Gardner TED loop bandwidth BnT
--timing_damping	0.707	Gardner TED damping
--phase_loop_bw	0.02	carrier phase PLL bandwidth
--phase_damping	0.707	carrier phase PLL damping

Equalizer

Option	Default	Controls
--eq_type	None	None / LMS / RLS / DFE (see §6 — LMS diverges OTA)
--eq_taps	11	equalizer tap count
--eq_mu	0.3	NLMS step (DD / real-preamble training)
--eq_dd	false	decision-directed tracking after LS training

RF hardware (TX)

Option	Default	Controls
--tx-args	""	UHD TX device (e.g. serial=30CD424)
--tx-rate	1.6e6	TX sample rate (Hz) = symbol_rate·U/D
--tx-freq	2.412e9	TX centre frequency (Hz)
--tx-gain	20	TX gain (dB)
--tx-bw	1.0e6	TX analog bandwidth (Hz)
--tx-ant	TX/RX	TX antenna port
--tx-subdev	A:A	TX subdev spec
--tx-channel	0	TX channel index

RF hardware (RX)

Option	Default	Controls
--rx-args	""	UHD RX device (e.g. serial=30CD3F7)
--rx-rate	1.6e6	RX sample rate (Hz), must equal TX rate
--rx-freq	2.412e9	RX centre frequency (Hz)
--rx-gain	30	RX gain (dB) — key for EVM/clipping (§ recipes)
--rx-bw	1.0e6	RX analog bandwidth (Hz)
--rx-ant	RX2	RX antenna port
--rx-subdev	A:A	RX subdev spec
--rx-channel	0	RX channel index

Common RF / buffering / misc

Option	Default	Controls
--ref	internal	clock reference: internal / external / mimo
--settling	0.2	RF settling time (s)
--uhd_timeout	1000	UHD TX timeout (ms)
--samps_per_buff	10000	samples per UHD receive buffer
--num_rcv_request	0	total samples to receive (0 = continuous)
--AGC_type	Feed	Feed (feed-forward) or Closed (closed-loop)
--dc-block	false	experimental RX DC-block high-pass (see §13)
--tx-dc-i / --tx-dc-q	0 / 0	manual TX LO-leakage null, normalized [−1,1] (§13)
--skip-rate-check	off	bypass startup rate-consistency check
--viz	true	capture + auto-save TX/RX figure on exit
--viz-dir	viz	base dir for figures (per-scheme subfolder made)

12. Command reference — per scheme (915 MHz, VERT900 antennas)

Copy-paste **sink** / **source** **ARQ** pairs with gains tuned per scheme from over-the-air testing (two B210s ~10 cm apart). All use stop-and-wait ARQ (TCP ACK on localhost) + FEC, and auto-save plots to `phy_outputs/<scheme>/figure.png`. The same set stands alone in the repo as `COMMANDS.md`, which also carries adapted (untested) command templates for the **N210** / **X310** / **X410** — only the device args, subdev, antenna and gain range change; the DSP flags are identical, and a shared 10 MHz reference (`--ref external`) on those platforms lifts the §13 dense-QAM limitation.

Rig: RX/sink serial 30CD3F7 (ant RX2), TX/source serial 30CD424 (ant TX/RX), both subdev A:A. Common flags: `--rx-freq/--tx-freq 915e6, --rx-rate/--tx-rate 1.6e6, --ack-transport tcp --ack-port 5599 --det-mult 3 --fec true`. **Start the sink (RX) first.**

Tuned gains per scheme (and why they differ)

Scheme	Waveform	--rx-gain	--tx-gain	extra	OTA status
BPSK	sc	20	78	—	very robust
QPSK	sc	20	78	—	solid (5/5)
8-PSK	sc	16	86	—	usable — a few ARQ retransmits (EVM ~16 %)
DBPSK	sc	20	78	—	solid (5/5)
DQPSK	sc	20	78	—	solid (5/5)
8-DPSK	sc	16	86	—	usable — a few ARQ retransmits
BPSK	ofdm	22	80	--ofdm-tx-peak 0.5	robust
QPSK	ofdm	22	85	--ofdm-tx-peak 0.5	solid

The pattern: **8-ary schemes need more TX power (86) and lower RX gain (16)** — a stronger signal buys the SNR their tighter decision regions demand without overdriving the front end. The key lever is **TX power, not RX gain** (raising TX 76->86 roughly halved 8-PSK EVM, 28 %->16 %). Higher-order QAM (16-QAM+) is omitted — the ~10 cm OTA link floors at ~28–31 % EVM, too noisy; those need the cabled link (below). Differential schemes run on the default `--eq_type None`; **do not combine differential with OFDM** (its pilots already handle phase).

Single-carrier (`--waveform sc`, default)

```
# QPSK - sc (robust tier: also BPSK / DBPSK / DQPSK, same gains)
# RX / sink
./sdr_system --role sink_arq --rx-args serial=30CD3F7 --tx-args serial=30CD3F7 \
  --rx-subdev A:A --rx-ant RX2 --rx-freq 915e6 --rx-rate 1.6e6 --rx-gain 20 \
  --scheme QPSK --fec true --ack-transport tcp --ack-port 5599 --det-mult 3
# TX / source
./sdr_system --role source_arq --tx-args serial=30CD424 --rx-args serial=30CD424 \
  --tx-subdev A:A --tx-ant TX/RX --tx-freq 915e6 --tx-rate 1.6e6 --tx-gain 78 \
  --scheme QPSK --fec true --ack-transport tcp --ack-host 127.0.0.1 --ack-port 5599 --timeout 3000
```

For **BPSK** / **DBPSK** / **DQPSK** on single-carrier, use the identical commands with the scheme name swapped (`--scheme BPSK` / `DBPSK` / `DQPSK`) — same 20 / 78 gains.

```
# 8-PSK - sc (8-ary tier: also 8-DPSK, same gains - more TX, lower RX)
# RX / sink
./sdr_system --role sink_arq --rx-args serial=30CD3F7 --tx-args serial=30CD3F7 \
  --rx-subdev A:A --rx-ant RX2 --rx-freq 915e6 --rx-rate 1.6e6 --rx-gain 16 \
  --scheme 8-PSK --fec true --ack-transport tcp --ack-port 5599 --det-mult 3
# TX / source
./sdr_system --role source_arq --tx-args serial=30CD424 --rx-args serial=30CD424 \
```

```
--tx-subdev A:A --tx-ant TX/RX --tx-freq 915e6 --tx-rate 1.6e6 --tx-gain 86 \
--scheme 8-PSK --fec true --ack-transport tcp --ack-host 127.0.0.1 --ack-port 5599 --timeout 3000
```

For **8-DPSK**, same commands with `--scheme 8-DPSK`.

OFDM (`--waveform ofdm`, 64 subcarriers, CP 16)

```
# QPSK - OFDM (BPSK: same, --scheme BPSK, --tx-gain 80)
# RX / sink
./sdr_system --role sink_arq --rx-args serial=30CD3F7 --tx-args serial=30CD3F7 \
--rx-subdev A:A --rx-ant RX2 --rx-freq 915e6 --rx-rate 1.6e6 --rx-gain 22 \
--waveform ofdm --ofdm-fft 64 --ofdm-cp 16 --scheme QPSK --fec true \
--ack-transport tcp --ack-port 5599 --det-mult 3
# TX / source
./sdr_system --role source_arq --tx-args serial=30CD424 --rx-args serial=30CD424 \
--tx-subdev A:A --tx-ant TX/RX --tx-freq 915e6 --tx-rate 1.6e6 --tx-gain 85 \
--waveform ofdm --ofdm-fft 64 --ofdm-cp 16 --ofdm-tx-peak 0.5 --scheme QPSK --fec true \
--ack-transport tcp --ack-host 127.0.0.1 --ack-port 5599 --timeout 3000
```

One-way (no ACK) instead of ARQ: use `--role rx / --role tx --tx-reps 20` and drop the `--ack-*` flags. Run from `build/` and prefix `./`.

Level-tuning cheatsheet (the #1 cause of “nothing decodes”)

- RX Peak= is **post-AGC PAPR**; ~1.2–1.4 is normal, *not* ADC clipping. Only lower `--rx-gain` if chunks never decode **and** the raw front end is saturated.
- No detections (`bursts=0`) -> signal too weak: raise `--tx-gain` (or `--rx-gain`), or lower `--det-mult`.
- Detections but CRC always fails -> check the constellation EVM in the figure; if the order is too high for the link SNR, drop a scheme or improve the link. **TX power** is the main EVM lever.
- Keep `--fec true` on marginal links; it closes the last few % of bit errors.

Higher-order QAM (16-QAM and up)

A clean channel is necessary but **not sufficient** on this two-radio rig — see §13. Over the air the ~28–31 % EVM floor blocks it; on a direct cable the SNR is excellent but a carrier-leakage / free-running-clock problem blocks it too. **Dense QAM needs a shared reference clock.**

13. Known limitation — dense QAM needs a shared reference clock

Validated ceiling on this rig: QPSK and 8-PSK (both single-carrier and OFDM, over the air and over a cable). **16-QAM and higher do not decode** on the two free-running B210s, and the cause is understood.

Symptom

Over a direct SMA cable the SNR is excellent — the QAM *amplitude* is perfect (clean concentric rings at the right radii) and QPSK decodes — yet 16-QAM fails on **both** waveforms: single-carrier shows a **phase-rotating ring**, OFDM a **blob at the origin**.

Root cause

The two B210s run on **independent TCXOs** (no shared clock), so there is a real carrier frequency offset whose estimate jitters ± 1200 Hz. (The default CFO estimator is now the least-squares phase-slope of §5.4-C, which ~halves that jitter to $\approx \pm 450$ Hz at 12 dB SNR — this *widens the margin* for QPSK/8-PSK and the DQPSK range test, but does **not** lift the wall below: the residual beat still rotates a 16-point constellation faster than the DD

PLL can track, so a shared reference clock remains required for dense QAM.) On the cable, the **TX carrier / LO leakage couples straight into the RX** (a ~40 dB spike near DC) and **beats at that CFO** — a slowly drifting near-DC tone. It (a) rotates the constellation, which QPSK's wide decision regions survive but 16-QAM's don't (and the decision-directed phase PLL can't lock 16 points), and (b) dominates the AGC, collapsing the OFDM constellation.

What we tried (and why each failed)

- **Static DC / per-burst mean removal** — the leakage is a *drifting tone*, not a static DC, so mean subtraction can't remove it.
- **RX DC-block high-pass** (`--dc-block`) — a gentle cutoff has no effect; an aggressive cutoff removes the leakage but **distorts the m-sequence preamble** (ACQ correlation peak 69->45) and breaks sync. Left in as an off-by-default option.
- **Manual TX LO-leakage null** (`--tx-dc-i / --tx-dc-q`, via `set_tx_dc_offset`) — only shifts the DC spike ~7 dB, and noisily, so the spike is mostly **RX self-mixing**, not TX feedthrough. The B210 can't self-calibrate either (no cal antenna -> `uhd_cal_tx_dc_offset` fails).

The fix

Feed a **common 10 MHz reference** into both radios' REF IN and run both with `--ref external` (a signal generator's 10 MHz output split two ways, a GPSDO, or an OctoClock). That drives **CFO ≈ 0** (no residual rotation -> dense QAM decodes) *and* turns the leakage into a **true static DC** (removed by the RX DC-offset correction / OFDM's nulled DC subcarrier). This is the standard requirement for coherent operation between separate SDRs.

Without a shared clock, the alternative is dedicated DSP — most promisingly **OFDM with leakage-robust sync** (AGC and CFO estimation that exclude the DC region; OFDM's scattered pilots already track residual CFO the WiFi way) or **pilot-aided single-carrier** — a substantial effort with no guarantee, versus a ~\$100 reference clock that is guaranteed.

Note: WiFi radios are *also* free-running and still do 256-/1024-QAM — via per-packet preamble CFO estimation + continuous pilot tracking (which we have) *and* factory-calibrated TX LO leakage (which the B210, uncalibrated on a direct cable, lacks). Closing that gap is exactly the DSP-vs-shared-clock trade-off above.

14. LoRa / CSS (chirp spread spectrum) — decodable receiver

A parallel waveform to `sc/ofdm`, added in `include/lora.hpp` with the radio path in `phy/src/main.cpp` (`--waveform lora`). Unlike the constellation pipeline (m-sequence preamble + RRC + PLL), CSS has its own chirp-based sync and demod, so it runs at the sample level via `transmit_samples / PHYSICAL_LAYER::capture_raw` (like the tone path), bypassing the ACQ threads.

Modulation

Spreading factor SF -> $N = 2^{\text{SF}}$ chips per symbol, SF bits/symbol. The base up-chirp is $u(n) = \exp(j \cdot 2\pi(n^2/2N - n/2))$. Symbol s in $[0, N)$ is the base up-chirp cyclically shifted by s : $x_s(n) = u((n+s) \bmod N)$.

Demodulation (the decodable receiver)

Dechirp (multiply by the conjugate base chirp = down-chirp) then FFT — the peak bin is the symbol. The FFT coherently integrates all N chips, giving the LoRa **processing gain**: validated to decode at **-10 dB SNR** (below the noise floor). Steps:

1. **Preamble detection** — slide and find `n_preamble` up-chirps that dechirp to the *same* bin with a **sharp** peak (peak/avg gated, so noise/constant junk can't false-lock).
2. **Fine timing** — brute-force the sample shift τ in $[0, N)$ that makes the SFD down-chirps decode cleanly; this pins the frame boundary and separates timing from CFO.
3. **CFO** — the aligned preamble up-chirp peak gives the integer carrier-frequency offset (bins); de-rotated before data demod.

4. **Sync-word check** — verify the 2 sync symbols (below); reject foreign frames.
5. **Data** — dechirp+FFT argmax per symbol -> symbols -> bits -> CRC/FEC framing.

Frame + sync word

[preamble: 8 base up-chirps][sync word: 2 up-chirp symbols][SFD: 2 down-chirps][data symbols]

The **sync word** is a 1-byte network id sent as 2 symbols (each nibble $\times$ 8, LoRa-style): `--lora-sync-word default 0x12 (=18) private`, 0x34 (=52) public. The receiver rejects frames whose sync symbols don't match — network isolation. (Before this, the scheme had *no* sync word; the preamble was plain up-chirps straight into the SFD.)

include/lora.hpp API

- `base_chirp(SF, down)` — base up/down chirp of N samples.
- `mod_symbol(s, SF, up, out) / demod_symbol[q](r, SF, ref[, q])` — one symbol; q is the peak/avg (sharpness) used to reject false locks.
- `bits_to_symbols / symbols_to_bits` — SF-bit packing.
- `sync_symbols(sync_word, SF, s1, s2)` — network id -> 2 symbols.
- `modulate(bits, SF, n_preamble=8, sync_word=0x12)` -> complex samples (full frame).
- `sync(r, SF, n_preamble, cfo_bins&, sync_word=0x12)` -> data-start sample index (-1 if no valid frame); fills the integer CFO.
- `demodulate(rx, SF, nbits, n_preamble=8, sync_word=0x12)` -> bits.

Options

`--waveform lora`, `--lora-sf (7-12)`, `--lora-sync-word (int; must match TX/RX)`. The raw jamming sweep is separate: `--message-type chirp` with `--chirp-bw/--chirp-sf/--chirp-down`.

Validation & status

`tools/lora_loopback_test.cpp (c++ -std=c++17 -O2 -Iinclude tools/lora_loopback_test.cpp -o /tmp/lora_test && /tmp/lora_test)` — ALL PASS: clean, +137-sample timing offset, AWGN to -10 dB, integer CFO, and sync-word rejection. The over-the-air `--waveform lora` link compiles and the Python API is wired, but is **pending hardware validation** (needs the radios). One-way only for now — folding LoRa into `sink_arq/source_arq` is the follow-up.